Shengxin Hong

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EDUCATION

Washington University in St. Louis **Saint Louis, MO**
PhD student in Computer Science Aug 2025 – Present

University of Detroit Mercy **Detroit, MI**
B.S. in Software Engineering Sep 2021 - Mar 2025
GPA: 3.8/4.0

PROJECT

Smart Kitchen: AI Agents for Real-Time Video Understanding and Behavior Detection Oct 2025 - Present
Washington University in St. Louis & **Google** (Joint Project) | *LangGraph, Fine-tuning, VLM*

- Designed a low-latency, real-time vision-language agent that monitors cooking process and issues real-time safety prompts for users with cognitive impairments. (Code: https://github.com/ShengxinHong/SmartKitchen)
- Implemented a **real-time video ingestion** and **short-term memory** pipeline for an edge VLM (**Qwen-VL-8b**), using a video encoder to detect scene shifts and adaptively increase keyframe extraction, and constructing a sliding live graph (short-term memory) from VLM-extracted entities with embedding similarity merging. Combined adaptive keyframes and live graph context to enable low-latency, context-consistent understanding of ongoing kitchen activities.
- Developed a **Video GraphRAG** module for **long-video memory** and retrieval that incrementally merges expired live graphs into a persistent video graph memory, performs semantic graph retrieval from user queries, enabling retrieval of only relevant clip on graph (no need to re-process the full video each time) and improving efficiency for downstream reasoning.
- Deployed an **on-device AI agent** on an overhead-camera, forming an end-to-end smart-kitchen pipeline; ran an experiment with 200 participants, achieving +11.5% Precision and +13.6% F1 over the strongest baseline, improving the reliability of real-time safety prompts for early cognitive decline.

OmniCellAgent – Towards AI Co-Scientists for Scientific Discovery Jul 2025– Oct 2025
Washington University in St. Louis | *Autogen, LangGraph, Reinforcement Learning, Full Stack*

- Built OmniCellAgent based on **Autogen** and **LangGraph**, the first multi-agent “AI co-scientist” (include Orchestrator, Omics Data Analysis, Search agents). This system autonomously generate and iteratively refine hypotheses, to simulate the process of conducting single-cell research like humans expert.
- Implemented a GAT-based **GraphRAG** pipeline over OmniCellTOSG and STaRK-Prime (129K entities, 8.1M relations) using Llama-3.1-7B, to retrieve disease-specific biomarkers and pathways within seconds.
- Delivered a production-grade web service using **Flask + Gunicorn** with **Neo4j/SQL** backends and **Dash/Plotly** visualizations, supporting concurrent user sessions via **AutoGen** async agent runtime and providing biomarker/pathway insights in seconds. With this platform, it reduces the time spent manually organizing and analyzing data by 20 hours per week.

Open-domain Question Answering Agent Based on Multi-tool Calling Mar 2025 – Jul 2025
Baidu Inc. | *ReAct Agent, LangChain, Multi-tool Calling*

- Built an open-domain QA agent on **Ernie-4.0** using a **ReAct**-based workflow to autonomously orchestrate 49 APIs and answer complex multi-subquery questions (2-5 sub-questions) in travel, weather, and finance.
- Implemented an end-to-end multi-turn agent loop (**parallel + dependency-aware tool calls**) with filtering and retry logic, cutting average tool calls per query from 5.7 to 3.6 and invalid invocations by 35%, reducing per-query tool/compute spend by ~37% and delivering material infra cost savings at scale.

- Designed a multi-path API recall and **ranking module** (BM25, ERNIE-**Embedding** semantic search, ERNIE-medium multi-label classifier), improving API selection precision@1 from 78% to 91% and token-level F1 from 0.79 to 0.86, reducing misrouted API calls and retry turns and saving ~$50K/year in compute and tool-invocation costs at current traffic.

Real-time POI Recommendation System Mar 2025 – Jul 2025
Kafka, Streaming Compute, OLTP & Data Warehouse/Lake, MLOps Pipeline, Visualization, ETL pipelines
- Built a production-style data stack with **Kafka** and **Structured Streaming** (online feature computation), **PostgreSQL/Snowflake** + **Parquet storage**, and **Airflow DAGs** to orchestrate synthetic data generation, streaming jobs, model training, batch prediction, and visualization of user behavior and feature importance.

Web Based Music Player Application Nov 2024 – Feb 2025
MERN, CI/CD, DevOps, Testing
- Built a Spotify-like platform using **MERN** stack, integrating **Deezer** API for music search and streaming with **RESTful** services.
- Implemented **CI/CD** via **GitHub Actions**, **Docker** containerization, and automated testing with **Cypress** and **Jest**, plus **CodeQL** for security.
- Delivered dynamic front-end and optimized back-end, supporting seamless user interactions.

PUBLICATION

[1]. **Shengxin Hong**, Liang Xiao, Xin Zhang, Jianxia Chen, ArgMed-Agents: Explainable Clinical Decision Reasoning with LLM Discussion via Argumentation Schemes, 2024 IEEE International Conference on Bioinformatics and Biomedicine (BIBM). **(40 Citations)**

[2]. **Shengxin Hong**, Cai, C, Du, S, Feng, H, Liu, S, Fan, X. (2025). “My Grade is Wrong!”: A Contestable AI Framework for Interactive Feedback in Evaluating Student Essays. Artificial Intelligence in Education. AIED 2025.

[3]. Huang D, Li H, Li W, **Shengxin Hong**, Zhang H, Dickson P, Zhan M, Miller JP, Cruchaga C, Province M, Chen Y, Payne P, Li F. OmniCellAgent: Towards AI Co-Scientists for Scientific Discovery in Precision Medicine. bioRxiv. 2025 Aug 2:2025.07.31.667797. (Submitted to Nature Machine Learning)

[4]. Cai, C., **Shengxin Hong**, Ma, M. et al. Analyzing the teaching and learning environments through student feedback at scale: a multi-agent LLMs framework. Education and Information Technologies (2025).

[5]. **Shengxin Hong**, Liang Xiao, and Jianxia Chen. An Interaction Model for Merging Multi-Agent Argumentation in Shared Clinical Decision Making.” 2023 IEEE International Conference on Bioinformatics and Biomedicine (BIBM). IEEE, 2023.

AWARDS & FELLOWSHIP

Graduate Research Fellowship **- 109,850$ per year**, WashU Aug 2025 – Jun 2026
Academic Excellence Award **- Top 0.1%**, University of Detroit Mercy Jun 2025

SKILL

Programming Languages: Python, SQL, Java, JavaScript, C++, TypeScript, HTML/CSS
Framework & Libraries: PyTorch, TensorFlow, LangChain, LangGraph, AutoGen, Transformers, Pandas, NumPy, Vue, PySpark, FastAPI, Node.js
Tools & Technologies: RAG, Reinforcement Learning, Fine-tuning (LoRA/PEFT), VLM, PowerBI, Gunicorn
Cloud / DevOps / Systems: Git, Docker, Kubernetes (K8s), AWS, Linux, Airflow, Kafka, MLflow, CUDA
Database: MongoDB, Neo4j, MySQL, PostgreSQL, Snowflake